

# Seismic Signal Classification Research Based on Multi-Attention Mechanism Residual Network

Chenghuan Wei  
Guangxi Normal University  
Guilin, China  
chenghuanxn@163.com

Tingting Wang  
Institute of Geophysics, China Earthquake Administration  
Beijing, China  
wangtt0303@163.com

Zhichao Feng  
Guangxi Normal University  
Guilin, China  
wy17826099054@163.com

Hanming Huang\*  
Guangxi Normal University  
Guilin, China  
\*huanghm@gxnu.edu.cn

Bohong Zheng  
Guangxi Normal University  
Guilin, China  
zhengbohong2022@163.com

**Abstract**—The accurate identification of seismic signals is of great significance for cleaning earthquake catalogs and earthquake early warning. Traditional identification methods require extracting features from the time domain waveform signals as inputs for classification algorithms. This paper proposes a novel method for seismic signals classification based on multi-attention mechanism residual network, which directly utilizes the waveform signals that have undergone simple processing as the inputs for the network. The network model uses deep residual neural network as its fundamental structure and integrates channel attention mechanism and spatial attention mechanism, which optimizes the network's attention to critical information within seismic waveform signals. Furthermore, global average importance pooling is applied in the network to broaden the pooling field and acquire more feature information. We select 94 natural earthquake events and 98 artificial explosion events occurred from 2010 to 2016 near the national capital region of China such as Shunyi district in Beijing and Sanhe city in Hebei, with a magnitude ranging from 1.5 to 2.8, as the research subjects. We use the five-fold cross validation method for classification. The experimental results demonstrate that the method achieves exceptionally high accuracy in distinguishing between natural earthquake events and artificial explosion events, with the highest accuracy of 98.83% and the average accuracy of 98.53%. Compared with the traditional Support Vector Machine (SVM) method, ResNet18 and Deep Residual SPP Non-local Efficient Shrinkage Network (RSNN), ours demonstrates superior classification performance.

**Keywords**—seismic signal classification; deep residual network; attention mechanism; global average importance pooling

## I. INTRODUCTION

With the rapid development of the digital era and the large-scale deployment of earthquake observation instruments globally, the field of seismology research has entered the era of big data. However, the data collected by seismic observation networks not only contains natural earthquake data but also

comprises non-natural earthquake data such as artificial explosions and nuclear detonations. Failure to process these data in a timely manner can impact seismological research and contaminate earthquake catalogs [1]. The identification of event source types from massive volumes of observed waveform data has remained a hot topic in the field of seismology in recent years. The pursuit of more accurate and faster earthquake classification algorithms possesses vital research significance in contemporary times.

Since the 1950s, numerous domestic and foreign researchers have been devoted to the study of seismic signal classification and have developed various methods. The traditional classification methods are to extract distinctive geophysical features from waveform data, such as main frequency [2], amplitude spectrum [3], P-wave/S-wave spectral ratio [4], and seismic phase characteristics [5], and use these feature information as the basis for classification. As machine learning has advanced, a vast array of scholars has combined machine learning with seismic classification research, using machine learning to learn richer feature information from seismic signals for classification [6][7]. With the continuous optimization and innovation of machine learning in the computer era, deep learning has emerged as a more complex and advanced variant of machine learning. Moreover, numerous researchers have repeatedly confirmed the feasibility of seismic signals classification based on deep learning. Li et al. [8] extracted time-frequency features from seismic waveform signals through the short-time Fourier transform, and used them as the inputs of a convolutional network to identify natural earthquakes and artificial blasting. Yonggang et al. [9] used the power spectrum of seismic waveforms to construct a dataset and utilized a network model based on residual structure for earthquake classification.

To enable the network model to learn more critical feature information, this paper enhances the deep residual network and proposes a novel network model for seismic signals

classification. By employing global average importance pooling and various attention mechanisms within residual structure, advanced abstract features of the waveforms are extracted to

enhance the classification capability between natural earthquake signals and artificial explosion signals.

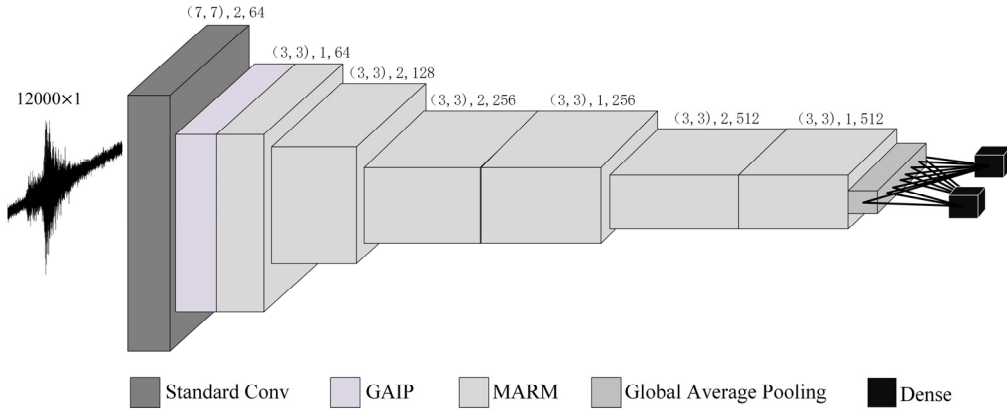


Figure 1. RGMN model structure

## II. THE INTRODUCTION OF RGMN MODEL

RGMN (Deep Residual GAIP Multi-attention Network) is based on deep residual neural network. It utilizes Global Average Importance Pooling (GAIP), and incorporates Multi-Attention Residual Module (MARM) which is formed by the integration of various attention mechanisms with residual module. The overall model of the network is shown in Fig. 1, which mainly consists of 1 convolutional layer, 1 global average importance pooling layer, 6 multi-attention residual modules, 1 global average pooling layer, and 1 fully connected layer.

The input size is  $12000 \times 1$  of the raw seismic signal. It undergoes a convolutional layer with a kernel size of  $7 \times 7$ , a stride of 2, and 64 channels. Subsequently, the data passes through a global average importance pooling layer to obtain a  $2999 \times 64$  pooled feature map. Multiple multi-attention residual modules are then applied to extract features from the pooled feature map, to fully mine various hidden information in the signal and obtain a feature map with a size of  $375 \times 512$ . Finally, a global average pooling layer followed by a fully connected layer produces the ultimate classification results of the network model.

### A. Global Average Importance Pooling

This paper introduces Global Average Importance Pooling (GAIP) as a replacement for max pooling. GAIP is an improvement upon the max pooling method, which is limited by the size of the pooling kernel and can only pool locally. In order to obtain a broader field of view and richer information through pooling, we incorporate global average pooling alongside max pooling. This permits us to obtain globally averaged important features, which are then broadcast-multiplied with the max pooling features. Seismic waveform signals portray a continuous process of variation, whereas max pooling fails to capture the overall change of the waveforms. Global average pooling can obtain continuous information of the entire waveform, and enables pooling to pay attention to the importance of the global average. We use a convolutional layer with a kernel size of  $1 \times 1$ ,

batch normalization layer, and activation functions Sigmoid( $\sigma$ ) and ReLU( $\delta$ ) to adjust the fusion of the two pooled features. The operation process is shown in Fig. 2.

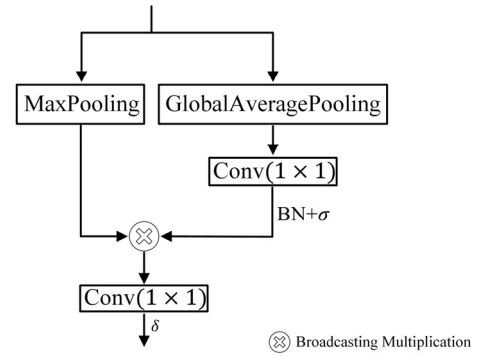


Figure 2. Global Average Importance Pooling

### B. Multi-Attention Residual Module

The waveform signals of natural earthquakes, often accompanied by considerable irrelevant noise, exhibit significant similarity to those of artificial explosions. A network model focusing solely on either the channel domain or the spatial domain will easily result in problems such as identification feature location errors and reduced classification accuracy.

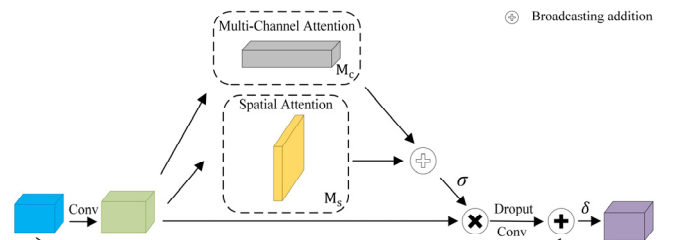


Figure 3. Multi-Attention Residual Module

Inspired by Convolutional Block Attention Module (CBAM) [10] and Attentional Feature Fusion (AFF) [11], we propose the Multi-Attention Residual Module (MARM). The overall structure of the module is shown in Fig. 3. The feature map  $M_c$  is obtained through multi-scale channel attention, and the feature map  $M_s$  is obtained through spatial attention. Subsequently, these two feature maps are added together after broadcasting, passed through Sigmoid activation, and then element-wise multiplied with the previous feature map. Finally, both attention mechanisms are integrated into the residual module.

The spatial attention module enables the network model to identify the location of critical information, as shown in Fig. 4. For the feature map  $M$  with dimensions  $W \times H \times C$  (width  $\times$  height  $\times$  number of channels), global average pooling and global max pooling are performed sequentially along the channel dimension. The resulting two pooled maps are concatenated, and after a convolution operation, the spatial attention feature map  $M_s$  is obtained with dimensions  $W \times H \times 1$ , as described in equation (1).

$$M_s = \text{Conv}[\text{Avgpool}(M), \text{Maxpool}(M)] \quad (1)$$

In the equation,  $\text{Conv}$  denotes the convolution operation,  $[\cdot]$  denotes feature maps concatenation.

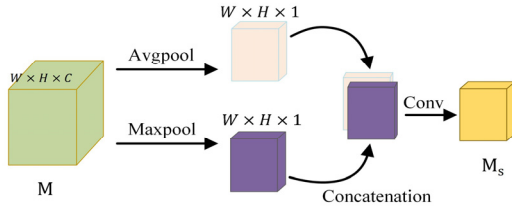


Figure 4. Spatial Attention Module

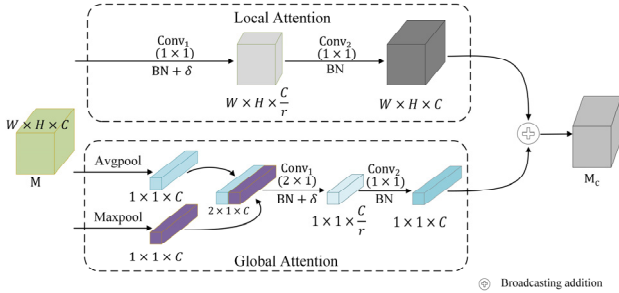


Figure 5. Multi-scale Channel Attention Module

The structure of the multi-scale channel attention module is shown in Fig. 5, which consists of global attention and local attention modules. The local attention module performs pointwise convolution on the input feature map with a kernel size of  $1 \times 1$  and  $\frac{C}{r}$  convolutional filters, and obtains new feature map with a size of  $W \times H \times \frac{C}{r}$ , which aggregates channel information. Subsequently, another convolution is applied to restore the original number of channels. The global attention module obtains global information from the input feature map by performing global average pooling and global max pooling along the spatial dimensions (width and height). This allows the feature map to retain average information while capturing

prominent features. Both pooling operations result in feature map of size  $1 \times 1 \times C$ , followed by convolution with a kernel size of  $2 \times 1$  for fusion and channel reduction. Similarly, then another convolution is performed to restore the number of channels. The feature maps obtained from two attention mechanisms are subjected to broadcasting addition to yield the multi-scale channel attention feature map ( $M_c$ ). The computational process is shown in equations (2)-(4).

$$M_l = \text{BN}(\text{Conv}_2(\delta(\text{BN}(\text{Conv}_1(M)))))) \quad (2)$$

$$M_g = \text{BN}(\text{Conv}_2(\delta(\text{BN}(\text{Conv}_1[\text{Avgpool}(M), \text{Maxpool}(M)])))) \quad (3)$$

$$M_c = M_l + M_g \quad (4)$$

$M$  denotes the input feature map,  $M_l$  and  $M_g$  respectively denote the feature maps from local attention and global attention mechanisms.  $\delta$  denotes activation function ReLU.  $\text{Conv}_1$  denotes channel reduction through pointwise convolution, and  $\text{Conv}_2$  denotes channel restoration through pointwise convolution.

The multi-attention module facilitates the residual module in gathering information from multiple attention fields and acquiring global and local information. Simultaneously, it empowers the network model to learn richer key features and further boosts the network's robustness and earthquake signals classification capability.

### III. EXPERIMENT AND RESULT

#### A. Data preprocessing and experimental setup

The experimental data includes 94 natural seismic events and 98 artificial explosion events that occurred from 2010 to 2016 near the national capital region of China such as Shunyi district in Beijing and Sanhe city in Hebei. The dataset contains waveform data of earthquake events in three components: east-west, north-south and vertical, alongside surface wave magnitude ranging from 1.5 to 2.8. By eliminating invalid waveforms that are overwhelmed by noise and fail to demonstrate the occurrence of seismic events, we have preserved a total of 7277 data records. Among these, 3770 records pertain to natural earthquake data, while 3507 records represent artificial explosion data.

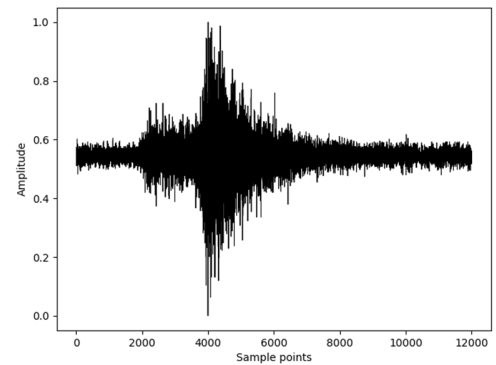


Figure 6. Natural earthquake waveform signal

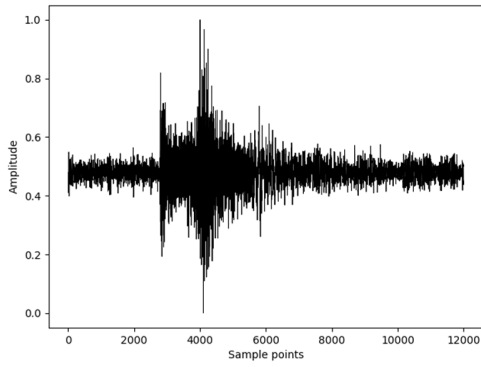


Figure 7. Artificial explosion waveform signal

Due to variations in the lengths of the original seismic waveform data, we truncate the data to 12000 samples, with a sampling frequency of 100 Hz. The truncated waveform segments encompass the occurrence of a single seismic event. To mitigate the impact of minor variations in magnitude on identification, we demean and normalize the data to the range of [0,1]. Fig.6 and Fig. 7 show the preprocessed waveform signal diagrams of natural earthquake and artificial explosion,

respectively. We use five-fold cross validation method to partition the preprocessed dataset into five equal subsets. Each subset contains approximately 754 natural earthquake waveform signals and 701 artificial explosion waveform signals. Four of them are designated as the training set, accounting for 80% of the total data, while the remaining subset serves as the test set, representing 20%. A different test set is selected each time. The division of dataset is shown in Table I.

Table I Division of natural earthquake and artificial explosion datasets

| Total data                           | Dataset division               | Type | Five rounds of training and test |      |      |      |      |
|--------------------------------------|--------------------------------|------|----------------------------------|------|------|------|------|
|                                      |                                |      | 1st                              | 2nd  | 3rd  | 4th  | 5th  |
| 7277<br>(eq:<br>3770<br>ep:<br>3507) | Training Set<br>(4/5)<br>≈5822 | eq   | 3016                             | 3016 | 3016 | 3016 | 3016 |
|                                      |                                | ep   | 2806                             | 2806 | 2806 | 2806 | 2804 |
|                                      | Test Set<br>(1/5)<br>≈1455     | eq   | 754                              | 754  | 754  | 754  | 754  |
|                                      |                                | ep   | 701                              | 701  | 701  | 701  | 703  |

The experiment is trained in the environment of NVIDIA GeForce GTX 1080Ti GPU. The training batch size is 64 and epoch is 1000. RMSprop optimizer is used for optimization. The flowchart of the method used in the paper is shown in Fig. 8.

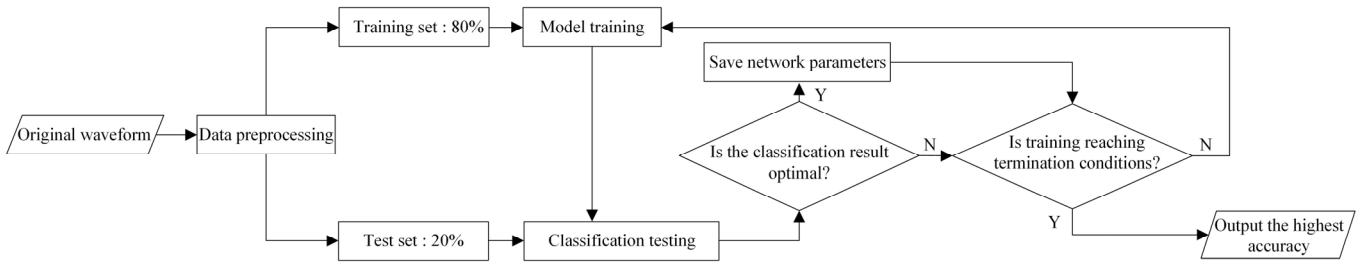


Figure 8. The flowchart of the method used in the paper

## B. Experimental result

To demonstrate the superior performance of RGMN proposed in this paper, we compare our model with Support Vector Machine (SVM), ResNet18 [12] and Deep Residual SPP Non-local Efficient Shrinkage Network (RSNN) [13]. The experimental results are shown in Table II.

Table II Comparison of accuracy (%) between the proposed method and other classification methods

| Test Frequency | Classification algorithm |          |       |             |
|----------------|--------------------------|----------|-------|-------------|
|                | SVM                      | ResNet18 | RSNN  | RGMN (ours) |
| 1st            | 82.96                    | 96.08    | 96.15 | 98.35       |
| 2nd            | 81.65                    | 95.60    | 96.84 | 98.49       |
| 3rd            | 82.99                    | 95.33    | 96.01 | 98.21       |
| 4th            | 82.47                    | 96.98    | 96.08 | 98.76       |
| 5th            | 81.19                    | 96.91    | 96.43 | 98.83       |
| Average        | 82.25                    | 96.18    | 96.30 | 98.53       |

By integrating multi-scale channel and spatial attention with residual module, our model is able to focus its attention on more crucial information from waveform data. This combination effectively addresses the shortcomings inherent in a single attention mechanism. Compared to the traditional residual network Resnet18, our model has a better classification effect, with an average classification accuracy that is 2.35% higher. Additionally, we adopt global average importance pooling instead of the traditional max pooling method to enhance the stability of network classification capability. This alteration expands the pooling receptive field from the size of local pooling kernels to encompass the entire global field, which allows the network to acquire more learnable information. The average classification accuracy of the SVM is 82.25%. SVM exhibits lower accuracy compared to our model. Additionally, the average classification accuracy of our model is 2.23% higher than that of the new model RSNN. The experimental results thoroughly demonstrate the effectiveness of our method for seismic signals classification.

#### IV. CONCLUSION

In this paper, we propose a method based on global average importance pooling and multi-attention residual network for the classification and recognition of natural earthquake signals and artificial explosion signals. In previous research, it has been customary to conduct feature extraction on waveform signals as a prerequisite. Contrastingly, in this paper, we construct the dataset by simply preprocessing the waveforms without performing extensive feature extraction. The features of waveform data are automatically extracted through deep learning algorithms. Such a data preprocessing method not only saves on manual processing costs but also provides valuable references for other signal research. Ultimately, our method achieves excellent classification results, with an average classification accuracy of 98.53%. It demonstrates excellent performance in the classification task of natural earthquake and artificial explosion signals, and further enhances our capabilities in earthquake recognition and early warning.

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